

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1697

Clodine 138 kV
December 22, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1697. The Interconnection Customer proposes a 65.3 MW battery energy storage facility interconnecting at the Clodine 138 kV substation in the Pennsylvania area. The request was received on March 20, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 444 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,628,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Elk Riv 138/1kV Auto	Eliza Crk / Ontelaunee Energy Ctr 345 KV	126.0	516	13.2
Ontelaunee Energy Center / Wyman-Gordon 138 KV	Ontelaunee Energy Center / Eliza Creek 345 KV	113.1	264	32.7
Kemmerer / Trisonic Substation 138 KV L	Wyman-Gordon / Ontelaunee Energy Center 138 KV L	111.3	185	60.0
Horizon Substation / Lake Louise Substation 138 KV	Mercury / Elk Riv 345 KV line	111.3	242	7.3
Trisonic / Wyman-Gordon 138 KV	Eliza Crk / Ontelaunee Energy Center 345 KV L	111.1	253	32.7
Redondo / Berthusen 138 KV line	Ontelaunee Energy Center / Eliza Creek 345 KV line	106.6	207	7.8
Gretna / Chama 138 KV Line	Wyman-Gordon / Ontelaunee Energy Center 138 KV line	105.8	237	60.0
Chama / South Reading 138 KV line	Wyman-Gordon / Ontelaunee Energy Center 138 KV	105.5	248	60.0
Horizon / Del Cerro 138 KV	Elk Riv / Mercury 345 KV line	105.1	231	7.3
Kemmerer / Gretna 138 KV	Wyman-Gordon / Ontelaunee Energy Ctr 138 KV Line	104.0	230	60.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Elk Riv 138 kV/1 kV Transformer	Replace transformer with higher-rated unit	\$5,958,000	26% (\$1,558,000)
Onelaunee Energy Center / Wyman-Gordon 138 KV	Rebuild line with higher-capacity conductor	\$2,834,000	100% (\$2,834,000)
Kemmerer / Trisonic 138 KV	Rebuild line with higher-capacity conductor	\$1,332,000	100% (\$1,332,000)
Horizon / Lake Louise 138 KV line	Rebuild line with higher-capacity conductor	\$1,514,000	100% (\$1,514,000)
Wyman-Gordon Substation / Trisonic 138 KV	Rebuild line with higher-capacity conductor	\$1,708,000	100% (\$1,708,000)
Redondo / Berthusen 138 KV line	Reconductor line	\$1,239,000	19% (\$236,000)
Gretna / Chama 138 KV Line	Reconductor line	\$534,000	100% (\$534,000)
Chama / South Reading 138 KV	Reconductor line	\$847,000	100% (\$847,000)
Del Cerro / Horizon 138 KV	Reconductor line	\$5,479,000	100% (\$5,479,000)
Kemmerer / Gretna 138 KV	Reconductor line	\$586,000	100% (\$586,000)
	Total allocated to project		\$16,628,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.